Lecture 1

$\mathrm{SO}(3)$, the double cover, and spherical harmonics

Objectives. *Same algebra, second group. We prove that the adjoint map is a two-to-one surjection* $\mathrm{SU}(2) \to \mathrm{SO}(3)$ *with kernel* $\{\pm\mathbb{1}\}$ *— the double cover — and read off its topology:* $\mathrm{SO}(3)$ *is the three-sphere with antipodal points identified, and it is not simply connected. We determine which spin representations descend (exactly the integer ones), completing Lecture 9's ledger; we realise them concretely as harmonic polynomials and spherical harmonics, proving* $\dim = 2\ell + 1$*, irreducibility, the Laplace eigenvalue* $-\ell(\ell+1)$ *and completeness in* $L^2(S^2)$*; we give Euler angles and Wigner matrices the single page they deserve; and we resolve Lecture 10's IOU: quantum mechanics acts on rays, Wigner's theorem makes that precise, and the physicist's rotation group is therefore* $\mathrm{SU}(2)$.

1.1 Two groups, one algebra

Lecture 8 computed the Lie algebra of the rotation group: $\mathfrak{so}(3)$, the antisymmetric 3×3 matrices, with basis $(L_a)_{jk} = -\epsilon_{ajk}$ and brackets $[L_a, L_b] = \epsilon_{abc} L_c$ — the same structure constants as $\mathfrak{su}(2)$. The map

$$\psi \colon \mathfrak{su}(2) \longrightarrow \mathfrak{so}(3), \qquad \psi(X_a) = L_a,$$

is a Lie algebra isomorphism (Lecture 8), and Lecture 9 unmasked it: $\psi = \mathrm{ad}$, the adjoint representation of $\mathfrak{su}(2)$ in the orthonormal basis $\{X_a\}$. Two consequences follow at once. First, the representation theory of the *algebra* $\mathfrak{so}(3)$ is already finished: transporting along ψ and complexifying (Lecture 10's lemma applies verbatim), the irreducible complex representations of $\mathfrak{so}(3)$ are the spin-j representations, one for every $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. Second, physics done infinitesimally cannot tell the two algebras apart: the Hermitian generators obey $[J_a, J_b] = \mathrm{i}\,\epsilon_{abc} J_c$ on either side of ψ. Whatever distinguishes $\mathrm{SO}(3)$ from $\mathrm{SU}(2)$ is invisible to commutation relations.

What distinguishes them is global, and Lecture 9 taught us its name. The ledger of integration (Table 9.1) has a blank row: is $\mathrm{SO}(3)$ simply connected, and which representations of its algebra integrate to the group? Today fills the row, and the instrument is the map ψ promoted from algebras to groups.

1.2 The double cover

Theorem 1.1 (The double cover)**.** *The adjoint map* $\Phi := \mathrm{Ad} \colon \mathrm{SU}(2) \to \mathrm{SO}(3)$*, written in the orthonormal basis* $\{X_a\}$ *as in Lecture 9, is a continuous surjective homomorphism with*

$$\ker \Phi = \{\pm\mathbb{1}\}.$$

Consequently Φ is exactly two-to-one — $\Phi(U) = \Phi(V)$ if and only if $V = \pm U$ — and Lecture 2's first isomorphism theorem gives

$$\mathrm{SO}(3) \;\cong\; \mathrm{SU}(2)/\{\pm\mathbb{1}\}. \tag{1.1}$$

Proof. That Φ is a continuous homomorphism into SO(3) is Lecture 9. Surjectivity: Lecture 9 also gives $\Phi(e^X) = e^{\mathrm{ad}_X} = e^{\psi(X)}$ for $X \in \mathfrak{su}(2)$; the isomorphism ψ is onto $\mathfrak{so}(3)$, and Lecture 8 proved the exponential map onto SO(3), so every rotation is $e^{\psi(X)} = \Phi(e^X)$. Kernel: $\Phi(U) = \mathbb{1}$ means $UX_aU^{-1} = X_a$, i.e. U commutes with each Pauli matrix. Commuting with σ_3 forces U to preserve the eigenlines of σ_3, hence to be diagonal, $U = \mathrm{diag}(a, d)$; commuting with σ_1, which swaps the two basis vectors, forces $a = d$; so $U = a\mathbb{1}$ with $a^2 = \det U = 1$, giving $U = \pm\mathbb{1}$. Finally $\Phi(U) = \Phi(V)$ iff $V^{-1}U \in \ker\Phi$ iff $U = \pm V$. ■

Remark 1.2 (A quotient that lands on its feet)**.** Equation (1.1) is the only place Part II touches an abstract quotient group. Quotients of matrix groups need not be matrix groups — nothing in Lecture 2's construction produces matrices — and our framework demands closed matrix groups. Here we are lucky twice over: the quotient is not merely isomorphic to some group, it is concretely realised, by Φ itself, as the matrix group SO(3). The abstract detour lasts one sentence.

Geometry first, then topology. Under the identification $\mathrm{SU}(2) \cong S^3$ of Lecture 10, the fibres of Φ are the pairs $\{\pm U\}$ — antipodal points of the sphere. So as a metric space, SO(3) *is the three-sphere with antipodal points glued*, each rotation owning exactly two antipodal addresses upstairs. The halved angles of Lectures 8–10 are this picture in coordinates (Figure 1.1): as θ runs from 0 to 2π, the rotation $R_z(\theta)$ walks a closed loop downstairs while its address $e^{-\mathrm{i}\theta\sigma_3/2}$ walks only half of a great circle upstairs, from $\mathbb{1}$ to $-\mathbb{1}$.

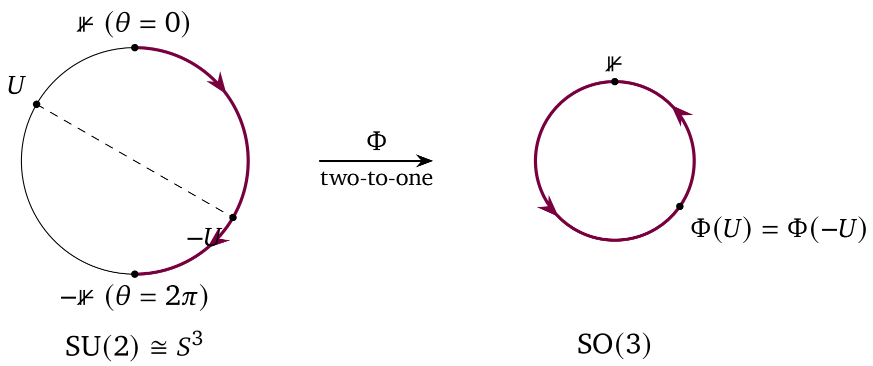

Figure 1.1: The double cover. Left: a great circle $\theta \mapsto e^{-\mathrm{i}\theta\sigma_3/2}$ of $\mathrm{SU}(2) \cong S^3$; as θ runs from 0 to 2π it traces the accented *open* path from $\mathbb{1}$ to $-\mathbb{1}$ — half the circle, the halved angle at work. Right: its image under Φ, the *closed* loop $R_z(\theta)$ in SO(3), traversed once in full. Antipodal points $\pm U$ (dashed chord) share a single image: Φ glues opposite points of the sphere. The open lift is the obstruction of Proposition 1.3 — contracting the loop downstairs would have to drag the lift's endpoint through the discrete fibre $\{\pm\mathbb{1}\}$ — while doubling the range to 4π closes the lift, which then contracts in S^3.

Proposition 1.3 (The loops of SO(3))**.** SO(3) *is connected (Lecture 8) but not simply connected: the loop $\rho(t) = R_z(2\pi t)$, $t \in [0, 1]$, is not contractible. Traversed* twice, *however, it is. (That* *every loop in* SO(3) *shares this fate — the obstruction is exactly a sign — we state without proof.)*

Proof (sketch; the patching follows Lecture 9). Doubling is the easy half. The loop ρ traversed twice is the Φ-image of $u(t) = e^{-2\pi \mathrm{i} t\, \sigma_3}$, which is a genuine loop in SU(2) ($u(0) = u(1) = \mathbb{1}$). Lecture 9 contracts u inside S^3; composing that homotopy with the continuous Φ contracts ρ^2 inside SO(3), basepoint fixed.

For the other half, lift. Around any point of SU(2) the map Φ is locally invertible — $\Phi(e^X) = e^{\psi(X)}$ together with Lecture 8's exponential charts shows it is a local homeomorphism with two continuous local inverses, differing by the sign of the sheet — so paths and homotopies in SO(3) lift to SU(2) by exactly the subdivide-and-patch argument that lifted circle loops in Lecture 9. The lift of ρ starting at $\mathbb{1}$ is $e^{-\mathrm{i}\pi t\, \sigma_3}$, ending at $-\mathbb{1}$: the loop opens up upstairs. If a homotopy contracted ρ, its lift would move the endpoint of the lifted path continuously through the discrete fibre $\{\pm\mathbb{1}\}$ — hence not at all — while connecting it to the lift of the constant loop, whose endpoint is $+\mathbb{1}$. Contradiction. ∎

Remark 1.4 (The belt trick)**.** Proposition 1.3 can be held in the hands. Give a belt a full 2π twist while keeping both ends fixed: no amount of manoeuvring the strap (= homotopy of the path of frames) removes the twist. Give it 4π, and the twist can be undone — Dirac's belt trick (Figure 1.2). Notebook 11 animates both, next to the corresponding paths in SU(2).

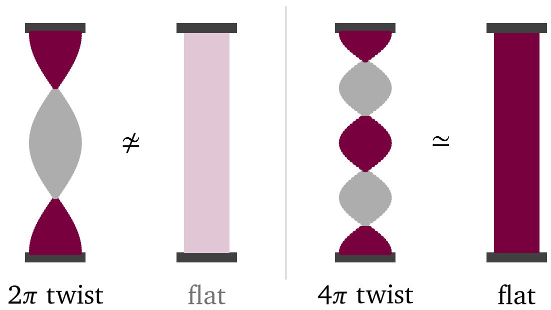

Figure 1.2: Dirac's belt trick, the topology of Proposition 1.3 held in the hands. A strap with both ends clamped records a path of frames by its twist; manoeuvring the strap without rotating the ends is a homotopy. A single 2π twist (left) cannot be removed — it is not homotopic to the untwisted strap, just as $\rho(t) = R_z(2\pi t)$ is not contractible in SO(3). A 4π twist (right) *can* be worked out flat by passing the strap over the clamped end, exactly as ρ^2 contracts. The front of the strap is accented, its back grey, so each half-twist shows as a colour flip pinched to an edge.

So SU(2) and SO(3) share an algebra but not their loops, and by Lecture 9 that is precisely the difference that representation theory feels.

1.3 Which representations descend

Theorem 1.5 (Descent through the kernel)**.** *Composition with Φ is a bijection between continuous finite-dimensional representations of* SO(3) *and those representations Π of* SU(2) *with $\Pi(-\mathbb{1}) = \mathbb{1}$, matching invariant subspaces and irreducibility. In particular, since $\Pi_j(-\mathbb{1}) =$*

$(-1)^{2j}\mathbb{1}$ *(Lecture 10), the irreducible representations of* SO(3) *are exactly the* integer-*spin ones,*

$$\Pi_\ell, \qquad \ell = 0, 1, 2, \ldots, \qquad \dim \Pi_\ell = 2\ell + 1 = 1, 3, 5, \ldots,$$

and no continuous representation of SO(3) *contains half-integer spin.*

Proof. If $\widetilde{\Pi}$ represents SO(3), then $\Pi = \widetilde{\Pi} \circ \Phi$ is a continuous representation of SU(2) killing $\ker \Phi$; since Φ is onto, the two have the same invariant subspaces, and Π determines $\widetilde{\Pi}$. Conversely, if $\Pi(-\mathbb{1}) = \mathbb{1}$ then Π is constant on the fibres $\{\pm U\}$, so there is a unique $\widetilde{\Pi}$ with $\widetilde{\Pi} \circ \Phi = \Pi$; it is a homomorphism because Φ is onto, and continuous because Φ has continuous local inverses (the local sections of Proposition 1.3's proof), so that locally $\widetilde{\Pi} = \Pi \circ s$. Lecture 10's list and sign $(-1)^{2j}$ finish the classification. ∎

Lecture 9's ledger closes: SO(3) is *not* simply connected, and the algebra representations that integrate to it are exactly those built from integer spins. The algebra could not see this — ψ is an isomorphism — the loop legislated. And the verdict agrees with analysis: Lecture 9 derived integer quantisation of orbital angular momentum from single-valuedness on a circle of rotations; today's theorem is the same fact read globally, since function actions are honest SO(3) representations and must therefore carry integer spin only. Where half-integers live we already know (Lecture 10); *why they are lawful* is Section 1.6. First, the integer representations deserve their concrete home.

1.4 Spherical harmonics: the integer representations, made of functions

Lecture 10 built every spin from polynomials in two complex variables. The integer spins admit a second construction, on the physicist's home ground: polynomials on $\mathbb{R}^3$, restricted to the sphere. Write (x, y, z) for coordinates on $\mathbb{R}^3$, $r = |x|$, and let

$$P_\ell = \big\{\text{homogeneous complex polynomials of degree } \ell \text{ in } x, y, z\big\}, \qquad \dim P_\ell = \tfrac{(\ell+1)(\ell+2)}{2},$$

(count the monomials $x^a y^b z^c$ with $a + b + c = \ell$). The function action $(\Pi(R)f)(x) = f(R^{-1}x)$ of Lecture 3 preserves degree and acts continuously: a representation of SO(3) on P_ℓ. One operator tells the spaces apart.

Lemma 1.6 (The Laplacian is a rotation invariant)**.** *For every $R \in \mathrm{O}(3)$ and smooth f, $\Delta(f \circ R^{-1}) = (\Delta f) \circ R^{-1}$.*

Proof. For a linear map A the chain rule gives $\partial_a(f \circ A) = \sum_b A_{ba}\,(\partial_b f) \circ A$, hence

$$\Delta(f \circ A) = \sum_{a,b,c} A_{ba}A_{ca}\,(\partial_b\partial_c f) \circ A = \sum_{b,c} (AA^{\mathsf{T}})_{bc}\,(\partial_b\partial_c f) \circ A,$$

and $AA^{\mathsf{T}} = \mathbb{1}$ for orthogonal A. ∎

Definition 1.7 (Harmonic polynomials)**.** $H_\ell = \{f \in P_\ell : \Delta f = 0\}$. By Lemma 1.6, H_ℓ is an invariant subspace of P_ℓ.

How big is it? The answer drops out of an inner product tailored to differentiation — the same trick as Lecture 10's Gaussian, in differential clothing. For $f = \sum_\alpha c_\alpha x^\alpha$ write $\bar{f}(\partial) = \sum_\alpha \bar{c}_\alpha\, \partial^\alpha$, and on each P_ℓ define

$$\langle f, g \rangle_\partial = \big(\bar{f}(\partial)\, g\big)(0). \tag{1.2}$$

Monomials are orthogonal with $\langle x^\alpha, x^\alpha\rangle_\partial = \alpha! > 0$, so (1.2) is an inner product; and since multiplying a polynomial by $r^2 = x^2 + y^2 + z^2$ corresponds, after the substitution $x \mapsto \partial$, to applying Δ, it obeys the adjunction

$$\left\langle r^2 f, g\right\rangle_\partial = \langle f, \Delta g\rangle_\partial\,, \qquad f \in P_{\ell-2},\ g \in P_\ell.$$

Proposition 1.8 (Dimension of H_ℓ). *$\Delta\colon P_\ell \to P_{\ell-2}$ is surjective, and*

$$P_\ell = H_\ell \oplus r^2 P_{\ell-2} \quad \text{(orthogonal for (1.2))}, \qquad \dim H_\ell = 2\ell + 1.$$

Iterating, $P_\ell = H_\ell \oplus r^2 H_{\ell-2} \oplus r^4 H_{\ell-4} \oplus \cdots$, so on the sphere ($r = 1$) every polynomial restricts to a sum of restricted harmonics.

Proof. If $h \in P_{\ell-2}$ is orthogonal to the range of Δ, the adjunction gives $\left\langle r^2 h, g\right\rangle_\partial = 0$ for all $g \in P_\ell$, so $r^2 h = 0$ and hence $h = 0$: Δ is surjective, and $\dim H_\ell = \dim P_\ell - \dim P_{\ell-2} = \frac{(\ell+1)(\ell+2)-\ell(\ell-1)}{2} = 2\ell + 1$. The adjunction also identifies $\ker\Delta$ with the orthogonal complement of $r^2 P_{\ell-2}$, giving the decomposition; iterate on the second summand. ∎

The dimension $2\ell + 1$ is no coincidence, and the ladder machinery proves it. Assemble Lecture 9's generators for the three axes into Hermitian operators and ladders,

$$L_z = -\mathrm{i}\,(x\partial_y - y\partial_x), \qquad L_\pm = L_x \pm \mathrm{i}L_y = \pm z\,(\partial_x \pm \mathrm{i}\,\partial_y) \mp (x \pm \mathrm{i}y)\,\partial_z,$$

which satisfy Lecture 10's ladder brackets $[L_z, L_\pm] = \pm L_\pm$, $[L_+, L_-] = 2L_z$ — same structure constants, same theorems.

Theorem 1.9 (H_ℓ is the spin-ℓ representation). *H_ℓ is irreducible, of spin ℓ, with highest-weight vector*

$$w_\ell = (x + \mathrm{i}y)^\ell.$$

Proof. First, w_ℓ lives in H_ℓ and is highest of weight ℓ:

$$\Delta(x + \mathrm{i}y)^\ell = \ell(\ell-1)(x+\mathrm{i}y)^{\ell-2}\,(1^2 + \mathrm{i}^2) = 0, \qquad L_z w_\ell = \ell\, w_\ell, \qquad L_+ w_\ell = 0,$$

the last two because $(\partial_x + \mathrm{i}\,\partial_y)w_\ell = \ell(x+\mathrm{i}y)^{\ell-1}(1+\mathrm{i}^2) = 0$ and $\partial_z w_\ell = 0$. Now run Lecture 10's descent inside H_ℓ from w_ℓ: the string formulas held in any representation (their derivation used only the brackets), so $v_k = L_-^k w_\ell$ satisfies $L_z v_k = (\ell - k)v_k$ and $L_+ v_k = k(2\ell - k + 1)v_{k-1}$; the nonzero v_k are independent, and if $v_{N+1} = 0 \neq v_N$ then $0 = (N+1)(2\ell - N)v_N$ forces $N = 2\ell$ exactly. The span of $v_0, \ldots, v_{2\ell}$ is therefore a $(2\ell+1)$-dimensional invariant subspace of H_ℓ, irreducible by the simple-spectrum ladder argument of Lecture 10's polynomial model — and by Proposition 1.8 it *is* H_ℓ. ∎

Definition 1.10 (Spherical harmonics). $Y_{\ell m}$ is the restriction to S^2 of the basis vector of H_ℓ of weight m, normalised in $L^2(S^2)$ and phased by the ladder convention of Lecture 10. Concretely: $Y_{\ell\ell} \propto (x + \mathrm{i}y)^\ell\big|_{S^2}$ and $Y_{\ell,m-1} \propto L_- Y_{\ell m}$.

For $\ell = 1$ the whole construction fits in one line:

$$w_1 = x + \mathrm{i}y, \qquad L_- w_1 = -2z, \qquad L_-^2\, w_1 = -2\,(x - \mathrm{i}y), \qquad L_-^3\, w_1 = 0,$$

so $Y_{1,\pm1} \propto (x \pm \mathrm{i}y)\big|_{S^2}$ and $Y_{1,0} \propto z\big|_{S^2}$ — the p-orbitals of chemistry, generated rather than tabulated. Sheet 11 runs $\ell = 2$ and compares with the standard tables. Note also, free of charge from homogeneity: under the parity map $x \mapsto -x$ (which commutes with rotations), H_ℓ has the definite sign $(-1)^\ell$ — a fact Lecture 12's selection rules will spend.

So far the sphere has been a place to restrict polynomials. The restriction in fact carries all the structure: a homogeneous polynomial vanishing on S^2 vanishes identically (homogeneity: $u(x) = r^\ell u(x/r)$), so H_ℓ maps bijectively onto

$$\mathcal{Y}_\ell := \left\{ u\big|_{S^2} : u \in H_\ell \right\} \subset L^2(S^2),$$

and the bijection intertwines the rotation actions. Two theorems make $\mathcal{Y}_\ell$ the centrepiece of mathematical physics: its elements diagonalise the sphere's Laplacian, and together they fill $L^2(S^2)$.

Definition 1.11 (The spherical Laplacian)**.** For g smooth on S^2, let $(Eg)(x) = g(x/|x|)$ be its degree-zero homogeneous extension, and define $\Delta_{S^2} g := \left(\Delta\, Eg\right)\big|_{S^2}$. (In spherical coordinates this is the familiar angular operator $\sin^{-1}\theta\, \partial_\theta(\sin\theta\, \partial_\theta) + \sin^{-2}\theta\, \partial_\varphi^2$; the verification is Notebook 11's.)

Theorem 1.12 (The eigenvalue $-\ell(\ell+1)$)**.** *Every $Y \in \mathcal{Y}_\ell$ satisfies*

$$\Delta_{S^2} Y = -\ell(\ell+1)\, Y. \tag{1.3}$$

Proof. Write $Y = u|_{S^2}$ with $u \in H_\ell$; homogeneity gives $u = r^\ell\, EY$. For h homogeneous of degree zero, Euler's identity ($x \cdot \nabla h = 0$; differentiate $h(tx) = h(x)$ at $t = 1$) kills the cross term in

$$\Delta(r^\ell h) = (\Delta r^\ell)\, h + 2\, \nabla r^\ell \cdot \nabla h + r^\ell\, \Delta h = \ell(\ell+1)\, r^{\ell-2}\, h + r^\ell\, \Delta h,$$

using $\partial_a r^\ell = \ell\, r^{\ell-2} x_a$, hence $\Delta r^\ell = \ell(\ell+1) r^{\ell-2}$. Applying this to $h = EY$ and $u \in H_\ell$: $0 = \Delta u = \ell(\ell+1)\, r^{\ell-2} EY + r^\ell\, \Delta(EY)$. Since $\Delta(EY)$ is homogeneous of degree -2, restricting to $r = 1$ gives (1.3). ∎

Remark 1.13 (Analysis meets algebra)**.** The eigenvalue is Lecture 10's Casimir at $j = \ell$. Indeed $\mathcal{Y}_\ell$ is the spin-ℓ representation, on which $L^2 = L_x^2 + L_y^2 + L_z^2$ acts as $\ell(\ell+1)\,\mathbb{1}$ (Lecture 10's proposition, whose proof used only brackets), so on every $\mathcal{Y}_\ell$ — and hence, by the completeness theorem below, on a dense subspace of $L^2(S^2)$ —

$$-\Delta_{S^2} = L_x^2 + L_y^2 + L_z^2.$$

The spherical Laplacian *is* the Casimir of the rotation group; the Laplace operator has been doing representation theory since the eighteenth century.

Theorem 1.14 (Spherical harmonics are complete)**.** *The spaces $\mathcal{Y}_\ell$ are mutually orthogonal in $L^2(S^2)$, and their closed span is everything:*

$$L^2(S^2) = \widehat{\bigoplus_{\ell \ge 0}} \mathcal{Y}_\ell, \tag{1.4}$$

so the functions $Y_{\ell m}$, $\ell \ge 0$, $-\ell \le m \le \ell$, are an orthonormal basis of $L^2(S^2)$.

Proof. The rotation action on $L^2(S^2)$ is unitary: the surface measure is rotation-invariant (substitution). Let $\mathcal{P}_N$ be the space of restrictions to S^2 of polynomials of degree $\le N$: finite-dimensional, invariant, and — by Proposition 1.8 with $r \equiv 1$ on the sphere — equal to $\sum_{\ell \le N} \mathcal{Y}_\ell$. On $\mathcal{P}_N$ we have an honest finite-dimensional unitary representation, so its generators are anti-Hermitian (Lecture 9), the Casimir L^2 is Hermitian, and the $\mathcal{Y}_\ell$ are its eigenspaces with the distinct eigenvalues $\ell(\ell+1)$: mutually orthogonal, with direct sum $\mathcal{P}_N$. Inside each $\mathcal{Y}_\ell$ the ladder proposition of Lecture 10 makes the $Y_{\ell m}$ orthonormal.

Density: the restricted polynomials form a subalgebra of $C(S^2)$ containing the constants, separating points (the coordinates do) and closed under conjugation, hence uniformly dense by Stone–Weierstrass — imported here from the analysis course — and uniform density implies L^2-density on a finite measure space, with $C(S^2)$ dense in $L^2(S^2)$ from functional analysis. An orthonormal family with dense span is a basis. ∎

The physics writes itself. A particle in a central potential, $H = -\frac{1}{2\mu}\Delta + V(r)$, commutes with rotations (Lemma 1.6); separating $\psi = f(r)\,Y_{\ell m}$ and using Theorem 1.12 in the same computation that proved it,

$$-\tfrac{1}{2\mu}\Big(f'' + \tfrac{2}{r}f'\Big) + \Big(V(r) + \tfrac{\ell(\ell+1)}{2\mu r^2}\Big)f = E\,f\,:$$

the centrifugal barrier of every quantum mechanics course is a Casimir eigenvalue. Each radial solution arrives $(2\ell+1)$-fold degenerate — Lecture 9's degeneracy doctrine with the dimensions now computed — and *only* integer ℓ occurs, by Theorem 1.5. Hydrogen's spectrum is degenerate beyond this, n^2 rather than $2\ell+1$: a starred mystery (a hidden symmetry generated by the Runge–Lenz vector enlarges SO(3) to SO(4); optional project).

1.5 Euler angles and Wigner matrices

Computations with rotations need coordinates, and three angles suffice.

Proposition 1.15 (Euler angles)**.** *Every $R \in \mathrm{SO}(3)$ can be written*

$$R = R_z(\alpha)\,R_y(\beta)\,R_z(\gamma), \qquad \alpha,\gamma \in [0,2\pi),\ \beta \in [0,\pi]. \tag{1.5}$$

Proof. The unit vector $R\hat{z}$ has polar angles (β,α) for some $\beta \in [0,\pi]$, $\alpha \in [0,2\pi)$ (Figure 1.3), and $R_z(\alpha)R_y(\beta)\hat{z}$ is exactly that vector. Hence $Q = \big(R_z(\alpha)R_y(\beta)\big)^{-1}R$ fixes $\hat{z}$, so it preserves the xy-plane, restricts there to an element of SO(2), and equals $R_z(\gamma)$ for some γ. (At $\beta = 0$ or π only $\alpha \pm \gamma$ is determined — the chart has a seam, known to every gimbal; Sheet 11 examines it.) ∎

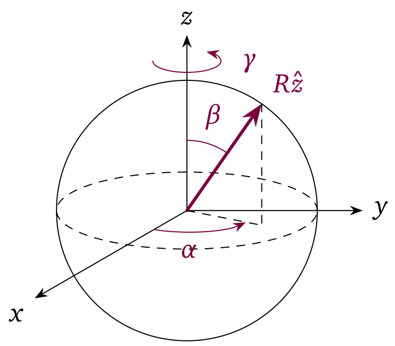

Figure 1.3: Euler angles. The pair (β,α) are the polar coordinates of $R\hat{z}$, the image of the north pole — β measured down from the z-axis, α around the equatorial plane from the x-axis (the dashed lines project $R\hat{z}$ onto that plane). These two angles fix $R_z(\alpha)R_y(\beta)$, which carries $\hat{z}$ to the right place; the leftover rotation about the original z-axis (upper circular arrow) is γ, the factor applied first in (1.5). At $\beta = 0$ or π the azimuth degenerates — the gimbal's seam.

Definition 1.16 (Wigner matrices)**.** $D^j_{m'm}(U) = \langle j, m' | \Pi_j(U) | j, m\rangle$, the matrix elements of the irreducible representations in the ladder basis. Each D^j is a unitary $(2j+1) \times (2j+1)$ matrix with $D^j(UV) = D^j(U)D^j(V)$; for integer j, Theorem 1.5 makes D^ℓ a function on SO(3) as well. In Euler coordinates, with $U(\alpha, \beta, \gamma) = e^{-\mathrm{i}\alpha J_z} e^{-\mathrm{i}\beta J_y} e^{-\mathrm{i}\gamma J_z}$,

$$D^j_{m'm}(\alpha, \beta, \gamma) = e^{-\mathrm{i}(\alpha m' + \gamma m)}\, d^{\,j}_{m'm}(\beta), \qquad d^{\,j}(\beta) = \text{matrix of } e^{-\mathrm{i}\beta J_y}, \tag{1.6}$$

so the whole j-dependence sits in the single real matrix $d^{\,j}$. For instance $d^{\,1/2}(\beta) = \begin{pmatrix} \cos\frac{\beta}{2} & -\sin\frac{\beta}{2} \\ \sin\frac{\beta}{2} & \cos\frac{\beta}{2} \end{pmatrix}$, half-angles included.

One identity ties the matrices to Section 1.4. In the basis $\{Y_{\ell m}\}$ of $\mathcal{Y}_\ell$ the function action has matrix elements D^ℓ: $\Pi(R)\, Y_{\ell m} = \sum_{m'} D^\ell_{m'm}(R)\, Y_{\ell m'}$. Only the $m' = 0$ harmonic survives at the north pole — from $\Pi(R_z(\theta)) Y_{\ell m'} = e^{-\mathrm{i}\theta m'} Y_{\ell m'}$ evaluated at the fixed point $\hat{z}$ — so evaluating the action there gives

$$D^\ell_{0m}(R) = c_\ell^{-1}\, Y_{\ell m}(R^{-1}\hat{z}), \qquad c_\ell = Y_{\ell 0}(\hat{z}) = \sqrt{\tfrac{2\ell+1}{4\pi}} \text{ (standard normalisation)} : \tag{1.7}$$

one row of every integer Wigner matrix is a spherical harmonic in disguise.

Example 1.17 (The spin-one Wigner matrix)**.** For $j = 1$ the exponential series collapses exactly as it did for the Pauli matrices in Lecture 8: the spin-one J_y of Lecture 10 has eigenvalues $0, \pm 1$, so $J_y^3 = J_y$ and the series folds into $e^{-\mathrm{i}\beta J_y} = \mathbb{1} - \mathrm{i}\sin\beta\, J_y + (\cos\beta - 1)\, J_y^2$, giving

$$d^{\,1}(\beta) = \begin{pmatrix} \frac{1+\cos\beta}{2} & -\frac{\sin\beta}{\sqrt{2}} & \frac{1-\cos\beta}{2} \\ \frac{\sin\beta}{\sqrt{2}} & \cos\beta & -\frac{\sin\beta}{\sqrt{2}} \\ \frac{1-\cos\beta}{2} & \frac{\sin\beta}{\sqrt{2}} & \frac{1+\cos\beta}{2} \end{pmatrix},$$

rows and columns ordered $m = 1, 0, -1$. Three readings. First, this is no stranger: it is the ordinary rotation $R_y(\beta)$ rewritten in the harmonic basis $\{-(x+\mathrm{i}y)/\sqrt{2},\ z,\ (x-\mathrm{i}y)/\sqrt{2}\}$ of H_1 — spin one is the vector representation (Sheet 10) in spherical dress. One column suffices to check: $\Pi(R_y(\beta))\, z = x\sin\beta + z\cos\beta$, whose coordinates in that basis are exactly the middle column. Second, the middle *row* is the identity (1.7) at $\ell = 1$: for instance $D^1_{00} = \cos\beta = c_1^{-1} Y_{10}(R^{-1}\hat{z})$, the polar angle of $R^{-1}\hat{z}$ being β; Sheet 11 checks the remaining entries, Euler phases included. Third, the endpoints: $d^{\,1}(\pi)$ is the antidiagonal flip $m \mapsto -m$ with signs $(-1)^{1-m}$, and

$$d^{\,1}(2\pi) = \mathbb{1}, \qquad \text{while} \qquad d^{\,1/2}(2\pi) = -\mathbb{1} :$$

Lecture 10's $(-1)^{2j}$, now wearing Euler coordinates.

Everything else about these matrices is computational and lives in Notebook 11 (`WignerD`) and on Sheet 11; Lecture 12 will need only their traces.

1.6 Rays, projective representations, and Wigner's theorem

One debt remains, and it is conceptual. Lecture 10 took the symmetry group of a quantum system to be SU(2) while geometry insisted on SO(3), and promised this lecture would arbitrate. The arbitration turns on what a quantum state actually is.

A state is not a vector. Two unit vectors differing by a phase, ψ and $e^{i\alpha}\psi$, are indistinguishable by any measurement: every prediction of quantum mechanics is built from transition probabilities

$$P\big([\varphi],[\psi]\big) = |\langle\varphi|\psi\rangle|^2\,,$$

which depend only on the *rays* $[\psi] = \{e^{i\alpha}\psi : \alpha\in\mathbb{R}\}$. The honest state space is the set of unit rays, and the honest notion of symmetry follows: a *quantum symmetry* is a bijection T of the rays preserving all transition probabilities. What such maps can be is settled by a theorem we import — the last conceptual import of the course.

Theorem 1.18 (Wigner's theorem; imported)**.** *Every quantum symmetry T is implemented by an operator $U\colon \mathcal{H}\to\mathcal{H}$ with $T[\psi] = [U\psi]$, where U is either unitary or* antiunitary *— additive, conjugate-linear, with $\langle U\varphi|U\psi\rangle = \overline{\langle\varphi|\psi\rangle}$ — and U is unique up to a phase.*

Reference for the omitted proof: Weinberg, *The Quantum Theory of Fields* I, Appendix 2.A.

Antiunitary implementers are not exotica — time reversal is one, as Sheet 7 noted — but they cannot occur in a connected symmetry group: every element of a connected matrix Lie group is a finite product of exponentials (Lecture 8), hence of squares $e^X = (e^{X/2})^2$, and the implementer of a square, $U_{h^2} = \omega\, U_h U_h$, is unitary whether U_h is unitary or antiunitary. For rotations, then, only unitaries — but unitaries defined up to phases, and the phases are the whole story.

Definition 1.19 (Projective representation)**.** A *projective unitary representation* of a group G on $\mathcal{H}$ is a map $g\mapsto U_g$ into the unitaries with

$$U_g\,U_h = \omega(g,h)\,U_{gh}, \qquad |\omega(g,h)| = 1\,: \tag{1.8}$$

a representation up to phases — exactly what Wigner's theorem delivers when a group acts by quantum symmetries.

Proposition 1.20 (Half-integer spin is projectively lawful)**.** *For every j, choose for each $R\in\mathrm{SO}(3)$ a preimage $U_R\in\Phi^{-1}(R)$ and set $W_R = \Pi_j(U_R)$. Then*

$$W_R\,W_{R'} = \pm\,W_{RR'}\,:$$

a projective unitary representation of SO(3)*, whose action on rays is independent of all choices. For integer j the signs can be removed (Theorem 1.5); for half-integer j they cannot:* no *honest representation of* SO(3) *induces the same ray action.*

Proof. Since $U_RU_{R'}\in\Phi^{-1}(RR') = \{\pm U_{RR'}\}$ and $\Pi_j(-U) = (-1)^{2j}\Pi_j(U)$ (Lecture 10), the multiplication rule holds with signs, and replacing U_R by $-U_R$ changes W_R by at most a sign — invisible on rays. Suppose now an honest representation $\widetilde{\Pi}$ of SO(3) induced the same ray action. For each $U\in\mathrm{SU}(2)$ the unitaries $\widetilde{\Pi}(\Phi(U))$ and $\Pi_j(U)$ implement the same ray map, so by the uniqueness in Wigner's theorem $\widetilde{\Pi}(\Phi(U)) = \lambda(U)\,\Pi_j(U)$ for a phase $\lambda(U)$; being a ratio of homomorphisms, λ is a continuous homomorphism $\mathrm{SU}(2)\to\mathrm{U}(1)$. Its derivative is a Lie algebra map $\mathfrak{su}(2)\to\mathfrak{u}(1)$ into an abelian algebra, and it must kill all brackets — but brackets span $\mathfrak{su}(2)$ ($X_1 = [X_2,X_3]$ and cyclically), so the derivative vanishes and $\lambda\equiv 1$ on the connected SU(2). Then $\widetilde{\Pi}\circ\Phi = \Pi_j$, forcing $\Pi_j(-\mathbb{1}) = \widetilde{\Pi}(\mathbb{1}) = \mathbb{1}$ — false for half-integer j. ∎

Here, at last, is the arbitration. Quantum mechanics is formulated on rays; its symmetry groups act by projective representations; and the projective representations of SO(3) are precisely the honest representations of SU(2) — the half we proved above, the converse (every projective representation of SO(3) arises so, after adjusting phases) being Bargmann's theorem, stated as Theorem 1.21 below and imported. *The physicist's rotation group is* SU(2), not by decree but because the quantum kinematics of rays makes the simply connected cover available free of charge. Lecture 10's IOU is paid in full: a spin-$\frac{1}{2}$ system violates nothing; it merely uses the phases that vectors carry and rays forgive. (Not every group is so lucky: for the Galilei group genuinely irremovable phases appear, carrying the particle's *mass* — close kin of the central element of Lecture 13's Heisenberg algebra. For the Lorentz group, Lecture 14 will find the same luck as here.)

Theorem 1.21 (Bargmann; imported)**.** *Let G be a connected Lie group whose Lie algebra $\mathfrak{g}$ has vanishing second cohomology, $H^2(\mathfrak{g}, \mathbb{R}) = 0$. Then every continuous projective unitary representation of G — every assignment $g \mapsto W_g$ obeying $W_g W_{g'} = c(g, g') W_{gg'}$ with phases $c(g, g') \in \mathrm{U}(1)$ — is, on rays, induced by an honest strongly continuous unitary representation of the universal cover $\widetilde{G}$: the phases $c(g, g')$ can be removed after lifting to $\widetilde{G}$. The hypothesis holds automatically when $\mathfrak{g}$ is semisimple — in particular for $\mathfrak{g} = \mathfrak{su}(2)$, by Whitehead's lemma $H^2(\mathfrak{g}, \mathbb{R}) = 0$.*

Reference for the omitted proof: Bargmann, *On unitary ray representations of continuous groups*, Ann. of Math. **59** (1954) 1–46; Weinberg, *The Quantum Theory of Fields I*, §2.7.

Applied to SO(3), whose universal cover is SU(2) with $\mathfrak{su}(2)$ semisimple, this supplies exactly the converse used above: every projective representation of SO(3) arises, after adjusting phases, from an honest representation of SU(2). (The Galilei group fails the hypothesis — $H^2 \neq 0$, the surviving cocycle being the mass — which is why its phases do not lift. For the connected Lorentz group the same argument uses the semisimple Lorentz algebra and lifts to its $\mathrm{SL}(2, \mathbb{C})$ cover, as Lecture 14 explains. The full Poincaré algebra is not semisimple; its distinct cohomological lifting input is stated at the point of use in Lecture 15.)

And the minus sign is not philosophy. In 1975 two independent crystal-neutron-interferometer experiments made it relative and therefore observable. Rauch and collaborators coherently split unpolarised slow neutrons, varied the unequal magnetic-field integrals along the two paths, and observed output-intensity oscillations with the predicted 4π spinor period. Werner and collaborators instead used a polarised beam and measured the interferometric phase shift produced by magnetic precession, directly identifying the sign reversal after a 2π rotation. Both experiments use magnetic precession and path recombination, but they are not the same protocol. The $(-1)^{2j}$ of Lecture 10, computed from bare algebra, is visible in a reactor hall.

Primary experimental sources: H. Rauch et al., "Verification of coherent spinor rotation of fermions," *Physics Letters A* **54** (1975), 425–427, doi:10.1016/0375-9601(75)90798-7; S. A. Werner et al., "Observation of the phase shift of a neutron due to precession in a magnetic field," *Physical Review Letters* **35** (1975), 1053–1055, doi:10.1103/PhysRevLett.35.1053.

The season's geography is complete: one algebra, two groups, a two-to-one bridge between them, and quantum mechanics choosing the upper deck. Lecture 12 restarts the engines of Part I — invariant integration in place of group sums, characters (including the crystal-field formula Part I borrowed on credit), and the question the experimentalists ask daily: what happens when two angular momenta meet. Lecture 14 will replay today's construction one signature up, with $\mathrm{SL}(2, \mathbb{C})$ standing over the Lorentz group.

Remark 1.22 (Computer algebra)**.** Notebook 11 checks `SphericalHarmonicY` and `WignerD` against our ladder constructions; verifies $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1) Y_{\ell m}$ numerically; plots $|Y_{\ell m}|$ over the sphere; and animates the double cover — a path of SU(2) matrices beside its image rotations, the 2π loop arriving at $-\mathbb{1}$ while the 4π loop closes — with the belt trick rendered on screen.

Summary.

- *Double cover.* $\Phi : \mathrm{SU}(2) \to \mathrm{SO}(3)$ is a two-to-one homomorphism with kernel $\{\pm\mathbb{1}\}$ (Theorem 1.1); SO(3) has fundamental group $\mathbb{Z}/2$ (Proposition 1.3).
- *Descent.* The representations of SO(3) are exactly the integer-spin representations of SU(2) (Theorem 1.5).
- *Spherical harmonics.* The degree-ℓ harmonic polynomials carry the spin-ℓ representation, of dimension $2\ell + 1$ (Theorem 1.9, Proposition 1.8), with spherical-Laplacian eigenvalue $-\ell(\ell+1)$ (Theorem 1.12).
- *Completeness.* The $Y_{\ell m}$ form an orthonormal basis of $L^2(S^2)$ (Theorem 1.14); Euler angles and the Wigner matrices coordinatise rotations (Proposition 1.15).
- *Projective representations.* Quantum states are rays (Wigner's theorem, Theorem 1.18), so half-integer spin is lawful and SU(2) is the operative group (Proposition 1.20).

Tutorial. Exercise Sheet 11.

References

Hall §4.7 (SO(3) versus SU(2)); Stillwell Ch. 2; Tung (the SO(3) and SU(2)–SO(3) chapters); Elliott–Dawber (rotation-group sections); Woit (rotations and spin); Costa–Fogli Ch. 2. For Bargmann's theorem (Theorem 1.21): Bargmann, Ann. of Math. **59** (1954) 1–46; Weinberg, *QFT I* §2.7. For the 1975 neutron experiments: Rauch et al., *Phys. Lett. A* **54**, 425–427; Werner et al., *Phys. Rev. Lett.* **35**, 1053–1055.

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